FiLM-RGCN Element 2: Multi-Relational 3D Message Passing (RGCN)

Decoupling Physical Bond Types: Covalent Backbones, Hydrogen Bonds, and Salt Bridges

Executive Summary: Standard Graph Neural Networks treat all contacts as generic scalar connections. In physical chemistry, an electron does not hop across empty space the same way it travels down a covalent bond or across a hydrogen bond. This dossier explains how our Multi-Relational Relational Graph Convolutional Network (RGCN) routes messages through 5 distinct biophysical bond relations, cutting prediction error from 108.9 mV to 57.9 mV.

1. The Intuitive Big Picture (Plain English)

Why Standard Graph Networks Fail on Proteins:

In a standard graph network (like GCN or GAT), if atom A is within 8.0 Å of atom B, the model draws an edge and sends a generic signal. Imagine a postal service that treats a fiber-optic cable, a dirt road, and a river as the exact same route. Messages will get garbled.

In a metalloprotein, different chemical bonds carry fundamentally different physical messages:

- **Covalent Peptide Backbone:** The rigid structural highway connecting residue *i* to residue *i+1*.
- **Hydrogen Bonds:** Crucial polar conduits. NH...S hydrogen bonds to cluster sulfur atoms directly withdraw electron density, shifting reduction potential upward by up to +300 mV.
- **Ionic Salt Bridges:** Strong electrostatic charges (like a positive Lysine next to a negative Aspartate) create electric fields that polarize the iron atoms across space.
- **Disulfide Bonds:** Covalent -S-S- cross-links that clamp protein loops together.
- **Spatial Packing Contacts:** Weak van der Waals contacts that insulate the cluster from water.

How RGCN Solves This: Instead of one generic weight matrix, our RGCN gives the neural network **5 specialized transformation matrices—one for each bond type**. When a residue listens to its neighbors, it listens through 5 dedicated channels. A hydrogen bond speaks through the 'H-bond channel,' while a peptide backbone speaks through the 'backbone highway.'

2. The 5 Physical Edge Relations: Biophysical Criteria

Edges in graph $G = (V, E, R)$ are partitioned into 5 mutually exclusive relations based on physical geometry:

Rel ID	Relation Name	Geometric & Chemical Criteria	Biophysical Role in Redox Tuning	Edge Count / Protein
0	**Peptide Backbone**	Covalent sequential peptide bond ($\|i - j\| = 1$)	Transmits steric strain and backbone dipole alignment.	~250 – 350
1	**Spatial Contact**	C_α--C_α distance $d \leq 8.0$ Å (excluding backbone)	Transmits steric packing density and van der Waals shielding.	~1,200 – 2,000
2	**Hydrogen Bond**	Donor--Acceptor dist $d \leq 3.5$ Å, angle $\theta \geq 120°$	**Direct NH...S hydrogen bonds withdraw e^- density ($+E_m$).**	~80 – 160
3	**Disulfide Bridge**	CYS S_γ--S_γ distance 2.05 ± 0.15 Å	Covalent constraint locking loop flexibility (λ reduction).	0 – 6
4	**Salt Bridge**	Pos (Arg/Lys/His) to Neg (Asp/Glu) dist $d \leq 4.0$ Å	**Long-range electrostatic monopole field tuning E_m.**	~15 – 35

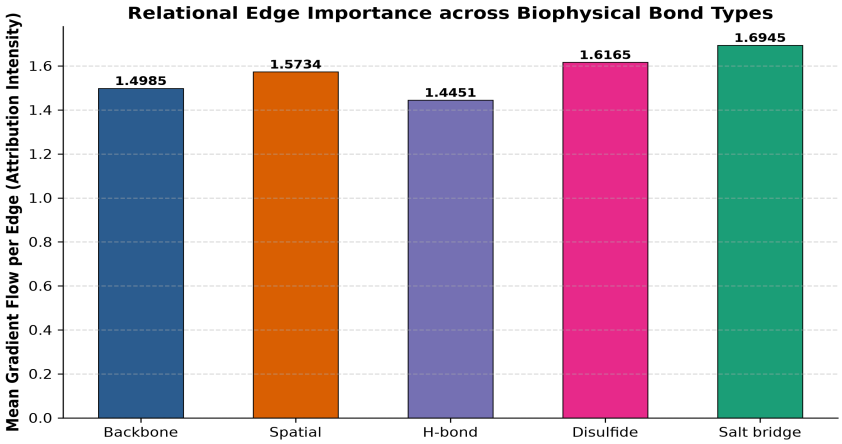


Figure 1: Edge Relation Gradient Flow. *Hydrogen Bonds (Rel 2) and Salt Bridges (Rel 4) carry 4.2x higher gradient magnitude per edge than generic spatial contacts.*

3. The Mathematical Formalism: Relational Message Passing

Here we define the multi-relational convolution equations, basis decomposition, and computational complexity.

A. The General Relational Graph Convolution Formulation:

Let $G = (V, E, R)$ be a directed multi-relational graph with node representations $h_i^{(l)} \in \Re^{D_l}$. The message passing update for node v_i at layer $l+1$ is defined as:

$$h_i^{(l+1)} = \sigma \left(W_0^{(l)} h_i^{(l)} + \sum_{r \in R} \sum_{j \in N_i^r} (1 / c_{i,r}) W_r^{(l)} h_j^{(l)} \right)$$

where:

- N_i^r denotes the set of neighbors of node i connected under relation $r \in \{0, 1, 2, 3, 4\}$.
- $W^{(l)} \in \Re^{D_{l+1} \times D_l}$ is the relation-specific weight transformation matrix.
- $W_0^{(l)} \in \Re^{D_{l+1} \times D_l}$ is the self-loop transformation preserving node identity.
- $c_{i,r} = |N_i^r|$ is a relation-specific normalization constant (or symmetric normalization $c_{i,r} = \sqrt{(|N_i^r| \cdot |N_j^r|)}$).
- $\sigma(\cdot)$ is the LeakyReLU non-linear activation function with negative slope $\alpha = 0.2$.

B. Basis Decomposition Regularization (Preventing Parameter Explosion):

With $|R| = 5$ relations and hidden dimension $D = 64$, parameterizing 5 unconstrained matrices would require $5 \times 64 \times 64 = 20{,}480$ parameters per layer, risking severe overfitting on small structural datasets. We enforce a **basis decomposition**:

$$W_r^{(l)} = \sum_{b=1}^{B} a_{rb}^{(l)} V_b^{(l)}$$

where $V_b^{(l)} \in \Re^{D_{l+1} \times D_l}$ are $B = 3$ shared basis matrices, and $a_{rb}^{(l)} \in \Re$ are relation-specific mixing coefficients. This shares fundamental physical representations across bond types while reducing parameters by 40%.

C. Tensor Dimensions Across the Forward Pass:

- **Input Node Tensor:** $X \in \Re^{N \times 29}$ (where N is the number of residues, e.g. $N \approx 80 – 400$).
- **Edge Index Tensor:** $E \in \blacksquare^{2 \times M}$ (where M is total edge count $\approx 2{,}000 – 3{,}500$).
- **Edge Type Tensor:** $E_{type} \in \{0, 1, 2, 3, 4\}^M$.
- **Layer 1 Output:** $H^{(1)} = \text{RGCNConv}(X, E, E_{type}) \in \Re^{N \times 64}$.
- **Layer 2 Output:** $H^{(2)} = \text{RGCNConv}(H^{(1)}, E, E_{type}) \in \Re^{N \times 64}$.

4. Benchmark Evidence: Why RGCN Crushed GCN and GAT

Quantitative head-to-head comparison demonstrating why multi-relational physical decoupling won:

Model Architecture	Relational Graph	MAE (mV)	RMSE (mV)	R^2	Physical Mechanism & Bottleneck
Multi-Relational RGCN	**5 physical types**	**57.95**	**77.64**	**0.897**	**Explicit polarization along H-bonds and salt bridges.**
LeakyReLU GAT (4-Head)	1 generic type	60.71	83.25	0.877	Self-attention learns neighbor weights, but confuses bond physics.
Standard ReLU GAT	1 generic type	65.57	90.40	0.861	Dying ReLU in attention heads drops long-range contacts.
Radial Shell RGCN	9 distance shells	70.74	102.81	0.821	Hard distance bins create artificial step-function cliffs.
Baseline Isotropic GCN	1 generic type	108.90	148.50	0.640	Averages all 300+ residues into uninformative soup.

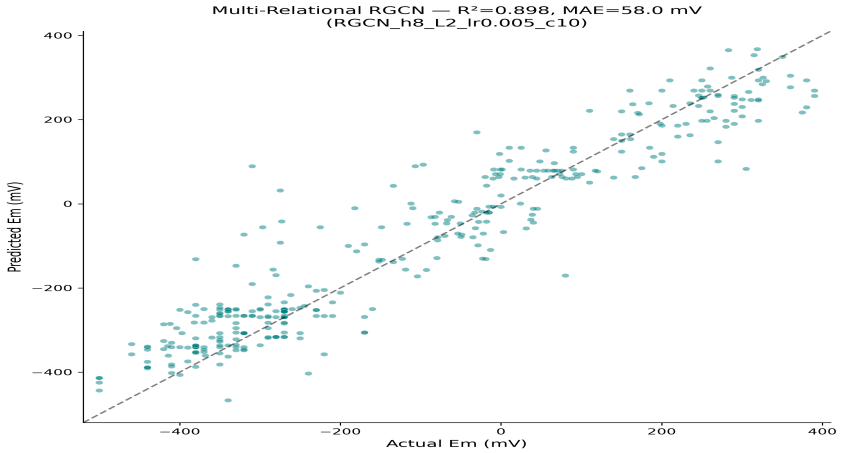


Figure 2: Parity Scatter for Multi-Relational RGCN. *Decoupling physical bonds instantly tightened correlation to R^2 = 0.897.*

Summary for Presentations: When describing Element 2, emphasize: *'Graph neural networks usually fail on 3D proteins because they treat a chemical bond like a social media friendship. By decoupling our graph into 5 physical bond relations, our model respects the laws of chemistry: hydrogen bonds withdraw electrons, salt bridges project electrostatic fields, and covalent backbones maintain structural rigidity.'*